

Zijing “Jimmy” Hu

Texas A&M University
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EDUCATION

Ph.D., Marketing, Texas A&M University, 2021-2027 (expected)
Committee: Yan “Lucy” Liu (Co-chair), Venkatesh Shankar (Co-chair), Lan Luo (USC), Bani Mallick (Statistics)
M.S., Business Analytics, Brandeis University, 2019-2021
B.Mgmt (Marketing), University of International Business and Economics, China, 2015-2019

RESEARCH INTERESTS

Substantive: New product management, Platform strategy, Pricing
Methodological: Causal inference, Machine learning, Structural economic modeling, Game theory

PAPERS UNDER REVIEW

1. **Zijing “Jimmy” Hu**, Venkatesh Shankar. From noise to insights: A deep learning model for identifying and measuring sales-driving product attributes. Revise & Resubmit, *Journal of Marketing Research*.
2. **Zijing “Jimmy” Hu**, Xuying Zhao. Rejection design for crowdsourcing platforms. Under Review, *Management Science*.

WORKING PAPERS

3. **Zijing “Jimmy” Hu**, Yan “Lucy” Liu. Discovery design in long-tail marketplaces: evidence from a digital music platform. *Job Market Paper*.
4. **Zijing “Jimmy” Hu**. Recovering true customer preferences through varying realization probabilities.

RESEARCH IN PROGRESS

5. Enhancing eco-driving behaviors using offline reinforcement learning (with Yuetao Gao, Shiyang Gong, Xiaomeng Du, and Yi Zhu).

RESEARCH PRESENTATIONS

*Early version

Discovery Design in Long-Tail Marketplaces: Evidence from a Digital Music Platform (with Yan “Lucy” Liu)	
Artificial Intelligence in Management Conference, Los Angeles, CA	Mar 2026
Rejection Design for Crowdsourcing Platforms (with Xuying Zhao)	
Platform Strategy Research Symposium, Boston University, Boston, MA	July 2025
Recovering True Customer Preferences Through Varying Realization Probabilities	
ISMS Marketing Science Conference, Washington D.C.	Jun 2025
UH Marketing Doctoral Symposium, University of Houston, Houston, TX	Apr 2025
Biz AI Conference, UT Dallas, Dallas, TX	Mar 2025
Artificial Intelligence in Management Conference, Los Angeles, CA	Mar 2025
Winter AMA Conference, Phoenix, AZ	Feb 2025
From Noise to Insights: A Deep Learning Model for Identifying and Measuring Sales-Driving Product Attributes (with Venkatesh Shankar)	
Theory + Practice in Marketing Conference, UT Austin, Austin, TX	May 2024
Artificial Intelligence in Management Conference, USC, Los Angeles, CA	Mar 2024
Conference on AI/ML/BA, Temple University, Philadelphia, PA	Dec 2023
*Artificial Intelligence in Management Conference, USC, Los Angeles, CA	Mar 2023
*ISMS Marketing Science Conference, virtual	Jun 2022

HONORS AND AWARDS

Graduate Fellow (working w/ Pradeep K. Chintagunta), Hagler Institute for Advanced Study, TAMU	2026-2027
Fellow, AMS Doctoral Consortium	2026
Google Cloud Research Credits	2025-2026
Dean's Award for Outstanding Teaching, Mays Business School, TAMU	2025
Student Presenter, UH Marketing Doctoral Symposium	2025
Texas Public Education Grant	2025, 2024
Fellow, INFORMS Marketing Science Doctoral Consortium	2025, 2024
Texas A&M University Graduate Fellowship	2021-2026

TEACHING

* Undergraduate-level, † Graduate-level

Instructor

Marketing Analytics*, Texas A&M University	Fall 2023
Dean's Award for Outstanding Teaching	

Teaching Assistant

Marketing Research*, Yan "Lucy" Liu, Texas A&M University	Spring 2026, 2024
Innovation and Product Management*, Rajan Varadarajan, Texas A&M University	Fall 2025, 2024
Marketing Metrics†, Yan "Lucy" Liu, Texas A&M University	Summer 2025, 2024
Marketing Research*, Rahul Suhag, Texas A&M University	Spring 2025
Marketing Analytics†, Xavi Vidal-Berastain, Brandeis University	Fall 2020
Machine Learning and Data Analysis†, Blake LeBaron, Brandeis University	Spring 2020

Guest Lecture

Strategic Digital Marketing*, Venkatesh Shankar, Texas A&M University	Fall 2022
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SERVICE

Organizer, Marketing Department New Doctoral Students Welcome Session	2024, 2025
Organizer, Marketing Department Brown Bag Seminar	2024

GRADUATE COURSEWORK

Marketing & Operations Management

Seminar in Marketing Models	Venkatesh Shankar
Seminar in Theory Construction & Knowledge Development	Manjit Yadav
Seminar in Marketing Strategy Research	Shrihari Sridhar
Seminar in Consumer Behavior	Keith Wilcox
Seminar in Platform Supply Chain Management	Xuying Zhao

Economics

Econometrics I	Qi Li
Econometrics II	Jason Lindo
Applied Microeconometrics	Yonghong An & Jackson Bunting
Game Theory	Silvana Krasteva
Industrial Organization I	Ben Klopach, Fernando Luco, & Steve Puller
Industrial Organization II	Ben Klopach, Fernando Luco, & Steve Puller
Experimental Economics	Catherine Eckel & Danila Serra
Causal Machine Learning (Audit)	Max Farrell & Sanjog Misra

Computer Science

Statistical Machine Learning	Luis Carvalho
Machine Learning Engineering	Tie Liu
Reinforcement Learning	Dileep Kalathil
Stochastic Systems (Audit)	P. R. Kumar

SKILLS

Programming & Software: Python, R, MATLAB, Stata, Mathematica, SQL
Languages: Mandarin Chinese (native), English (proficient)

REFERENCES

Yan “Lucy” Liu (co-chair)	Venkatesh Shankar (co-chair)	Xuying Zhao (co-author)
Professor of Marketing	Professor of Marketing	Professor of Operations Management
Mays Business School, TAMU	Cox School of Business, SMU	Mays Business School, TAMU
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SELECTED RESEARCH ABSTRACTS

Discovery design in long-tail marketplaces: Evidence from a digital music platform

Abstract: Digital platforms shape demand not only by matching consumers to products they already know, but also by determining which products consumers encounter in the first place. This paper studies how platform discovery architecture affects demand in a long-tail digital market. We distinguish between two channels of visibility: global exposure, generated by platform-wide signals, and local exposure, generated by the activity of socially proximate users in a network. We first document that discovery is rare but economically central: only a small share of transactions represents initial consumer-artist discovery, yet these discoveries generate substantial short-run follow-on purchases and long-run repeat demand. We then show that both global and local exposure increase subsequent sales, including sales from consumers who are distant from the artist in the network. To quantify and interpret these patterns, we develop a continuous-time encounter-choice model that separates exposure-driven consideration from preference-driven conversion. Counterfactual simulations show how market outcomes shift when these discovery channels are removed or redesigned. The results provide implications for platform design by showing how the allocation of visibility shapes the demand in long-tail markets.

From noise to insights: A deep learning model for identifying and measuring sales-driving product attributes

Abstract: Identifying sales-driving product attributes (that are not easily observable) and quantifying their impact on customer preference and sales is crucial for product development. Accomplishing these tasks with primary data is often costly and time-consuming. The use of secondary data, such as product reviews and ratings, is a viable option. However, these data are noisy and are often biased (e.g., fake reviews). Furthermore, results from most extant methods lack interpretability and explainability. We address these issues and propose a novel and integrative machine learning model, Non-Latent Variational Autoencoder (NLVAE), to recover critical but potentially unobservable product attributes from both structured and unstructured data. By embedding measurement error correction within the Variational Autoencoder framework, NLVAE not only accurately recovers product attribute scores but also offers enhanced interpretability and explainability. We theoretically derive the model’s properties and validate them using simulations. We apply NLVAE to a video game dataset, using a large language model (LLM) to analyze reviews, identify the underlying attributes, and debias and link them to sales to identify those that drive sales. We also show that product attribute scores extracted through the NLVAE model predict sales better than those from alternative models. We outline the implications for product development.

Rejection design for crowdsourcing platforms

Abstract: While crowdsourcing offers a scalable approach to harnessing collective intelligence, en-

sure submission quality remains a persistent challenge due to the variability of worker effort. To address this, many platforms empower requesters to reject unsatisfactory submissions with full refunds. However, industry practices of rejection governance vary significantly, ranging from unrestricted rejection rights to strict minimum acceptance quotas. The optimal design of the rejection policy remains underexplored. We develop a game-theoretic model, micro-founded on the optimal submission behavior of heterogeneous crowdworkers, to examine the strategic interactions among a platform, a requester, and the crowd. Our analysis yields several insights. First, both prohibiting rejection and allowing unrestricted rejection can be detrimental. These policies not only reduce crowdworker welfare and platform profit but also shrink the profitable market boundary, highlighting the indispensable role of active platform governance. Second, we characterize the platform’s optimal minimum acceptance restriction and show that it aligns requester incentives with platform objectives, achieving the first-best performance of a vertically integrated entity. Third, we show an equivalence between imposing a minimum acceptance requirement and charging for rejection through an explicit pricing rule. This equivalence clarifies the mechanism behind the policy and provides an alternative design that platforms can implement. Overall, our analysis provides guidance on how platforms should design rejection and refund rules to balance quality control, market participation, and welfare.

Recovering true customer preferences through varying realization probabilities

Abstract: Incentive-aligned choice experiments have emerged as a cost-effective solution for understanding customer preferences for products, given their external validity compared to stated-preference surveys and cost advantages over establishing a test market. Despite their advantages, recent findings suggest that these experiments may not fully capture true customer preferences. In this study, we introduce a novel approach to improve the external validity of incentive-aligned choice experiments. Our model integrates a measurement error framework with machine learning techniques. We theoretically demonstrate that this model ensures the identification of true customer preferences by leveraging variations in realization probabilities under realistic assumptions. We further validate our model through a series of simulations, effectively illustrating its capability to accurately recover true market share and granular patterns of customer behavior. This research equips both practitioners and scholars with a robust method for enhancing the application of incentive-aligned designs in market research.